As the voltage increases above rated value, when the input frequency goes above base frequency, only constant (rated) voltage with variable frequency (frequency control) is used for speed control. Under this condition, both flux and maximum torque decrease as the frequency is increased.

## PROCEDURE

Make the connections as shown in diagram.

Precautions

- i. TPST in open position
- ii. $DPST_1$ and $DPST_2$ in open position
- iii. Motor field rheostat in minimum position
- iv. Potential divider in minimum voltage position
- v. Autotransformer at minimum voltage position
- vi. Keep the belt on the brake drum of induction motor in loose position (induction motor on no load).

Switch on the DC supply to the DC motor by closing the switch $DPST_1$. Start the DC shunt motor using 3-point starter. Increase the resistance of dc motor field rheostat and drive the alternator at rated speed (1500rpm). Now, dc supply is given to the alternator field winding and adjusts the potential divider so that the generated voltage is rated value (400V). Close the TPST switch.

Increase the autotransformer. Induction motor starts running on no load. Apply rated voltage by adjusting autotransformer. Note down the frequency, voltage and speed of the induction motor. Now, increase the frequency keeping the voltage constant (=400V). Again, note down frequency, voltage and speed each time. Repeat the procedure till frequency reaches 54Hz.

Now, decrease the frequency till it becomes 50Hz. Decrease the voltage and frequency in proportion ($\frac{V}{f} = \frac{400}{50} = 8$) and note down the frequency, voltage and speed of the induction motor each time. This procedure is continued till frequency decreases to 44Hz.

Repeat the above procedure for another load (say $1_L = 4A$). Switch off the supply after bringing the motor to no-load.